



Ministry of Jal Shakti  
Department of Water Resources,  
River Development & Ganga Rejuvenation  
Government of India

## Annual Conference of RC/Secy(DM)/SDRF/Civil Defense, etc. - 2026

# Flood Preparedness of CWC



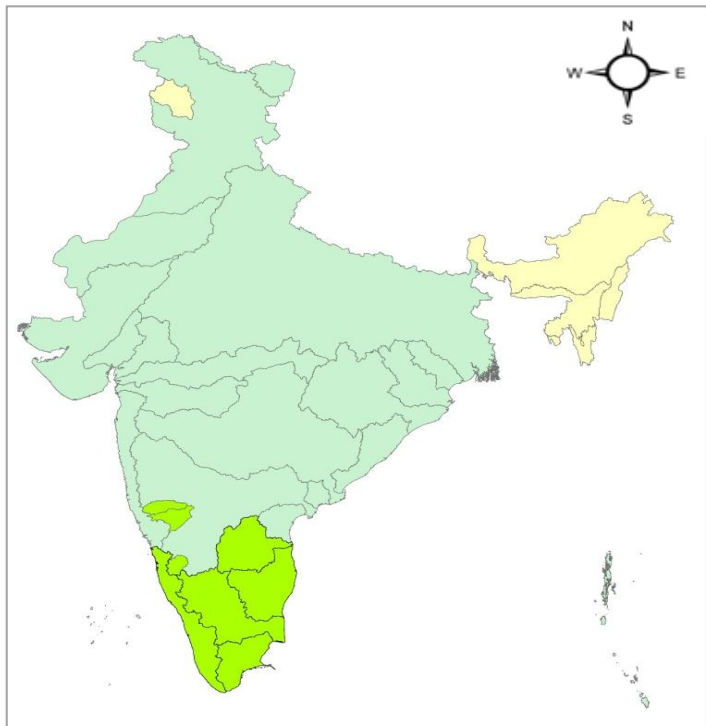
02.05.2026



CENTRAL WATER  
COMMISSION

# Flood Preparedness by CWC

## Designated Flood Period and categories



**Predefined Flood Periods are fixed based on regional rainfall distribution and the climatological onset and withdrawal of monsoon**

**1<sup>st</sup> May to 31<sup>st</sup> October** [Brahmaputra & Barak (including Teesta, Rivers flowing in States of Tripura, Manipur, Mizoram, Nagaland, Meghalaya, North Bengal, Sikkim) and Jhelum Sub Basin of Indus.]

**1<sup>st</sup> June to 31<sup>st</sup> October** [All other Basins up to Krishna Basin excluding the sub basins of Krishna such as Ghataprabha, Malaprabha and sub basins of Tungabhadra such as Tunga and Bhadra]

**1<sup>st</sup> June to 31<sup>st</sup> December** [All Basins South of Krishna Basin and the sub basins of Krishna such as Ghataprabha, Malaprabha, and sub basins of Tungabhadra such as Tunga and Bhadra]

### FLOOD CATEGORIES

Highest Flood Level

EXTREME

Danger Level

SEVERE

Warning Level

Above NORMAL

### ALERT COMMUNICATION PROTOCOLS

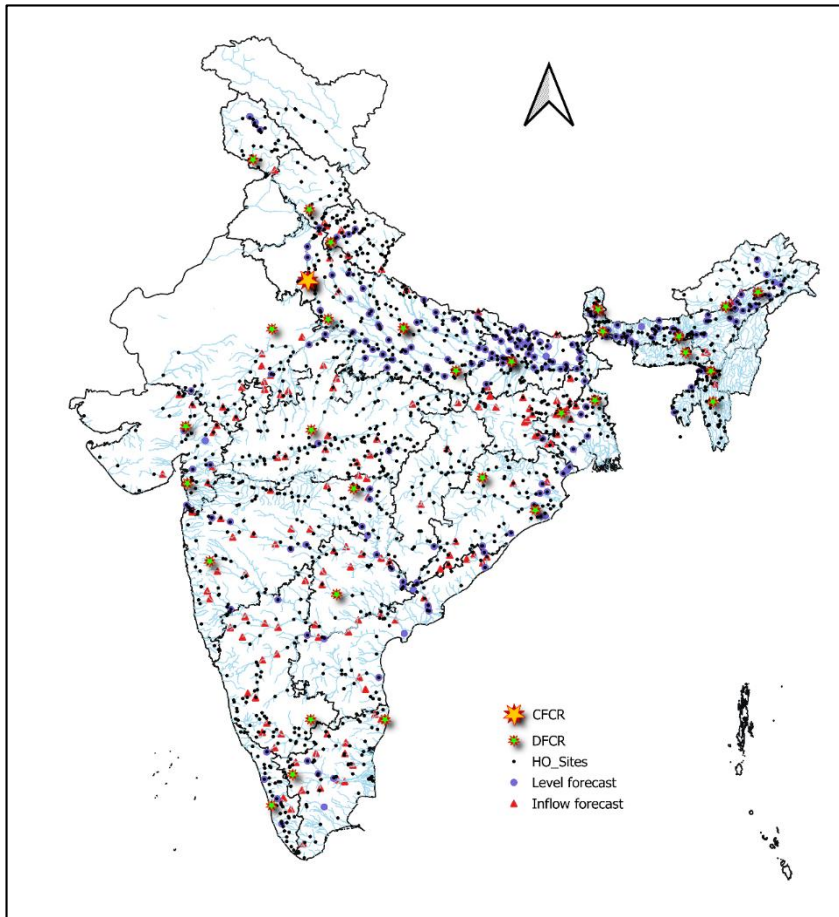
**Red Stage** 3 hourly or more frequent updates

**Orange stage** : 6 hourly updates

**Yellow stage** : daily updates

# Flood Forecasting Activities

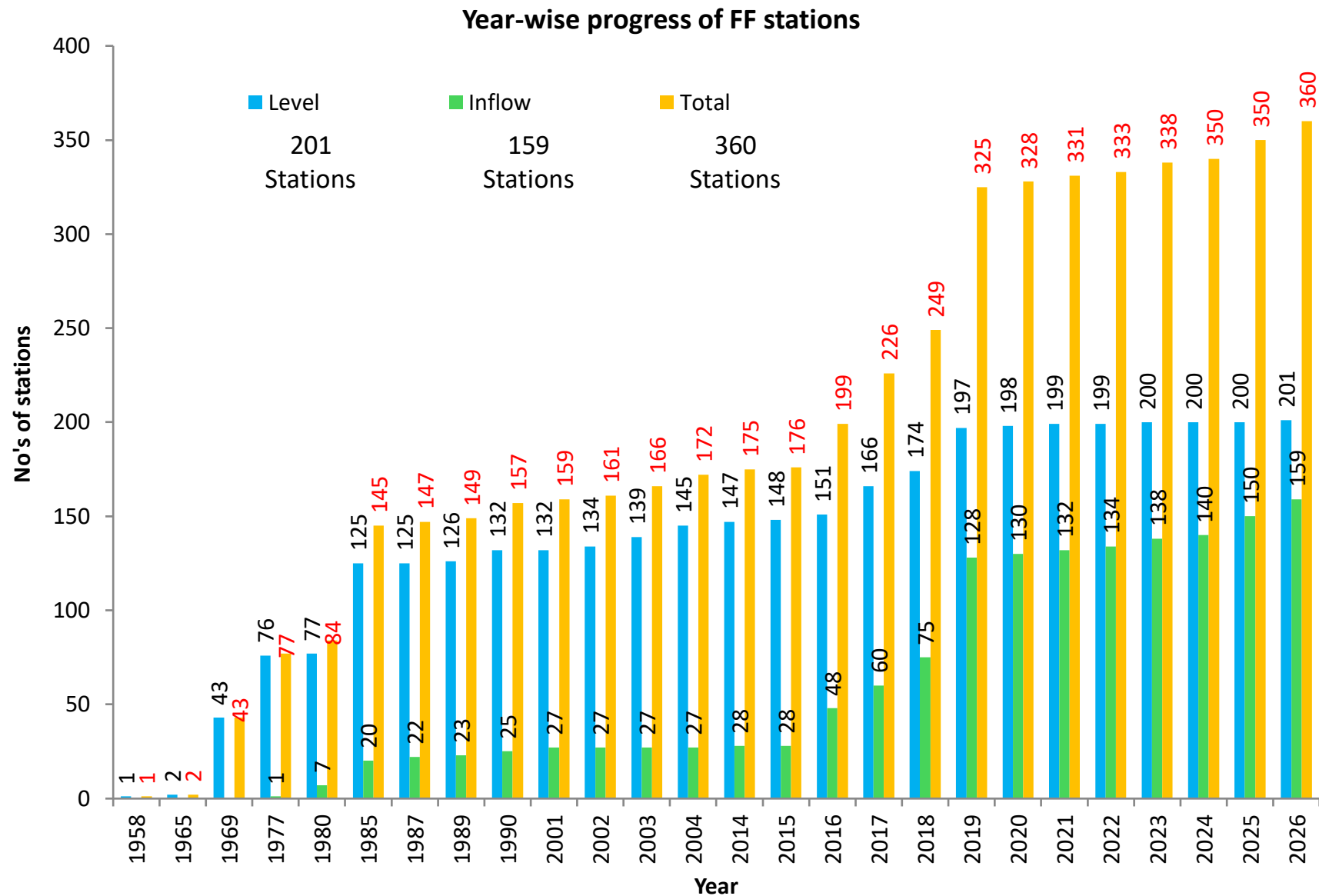
## Hydrological Station network, Flood Modelling and Forecasting



- 1543 Hydrological Observation Stations
- 360 Flood Forecasting Stations  
(201 Level + 159 inflow)
- 3hr/6 hr/12hr/daily/7 days – updates  
/forecast/ advisory
- 1+36 Control Rooms

# Flood Preparedness by CWC

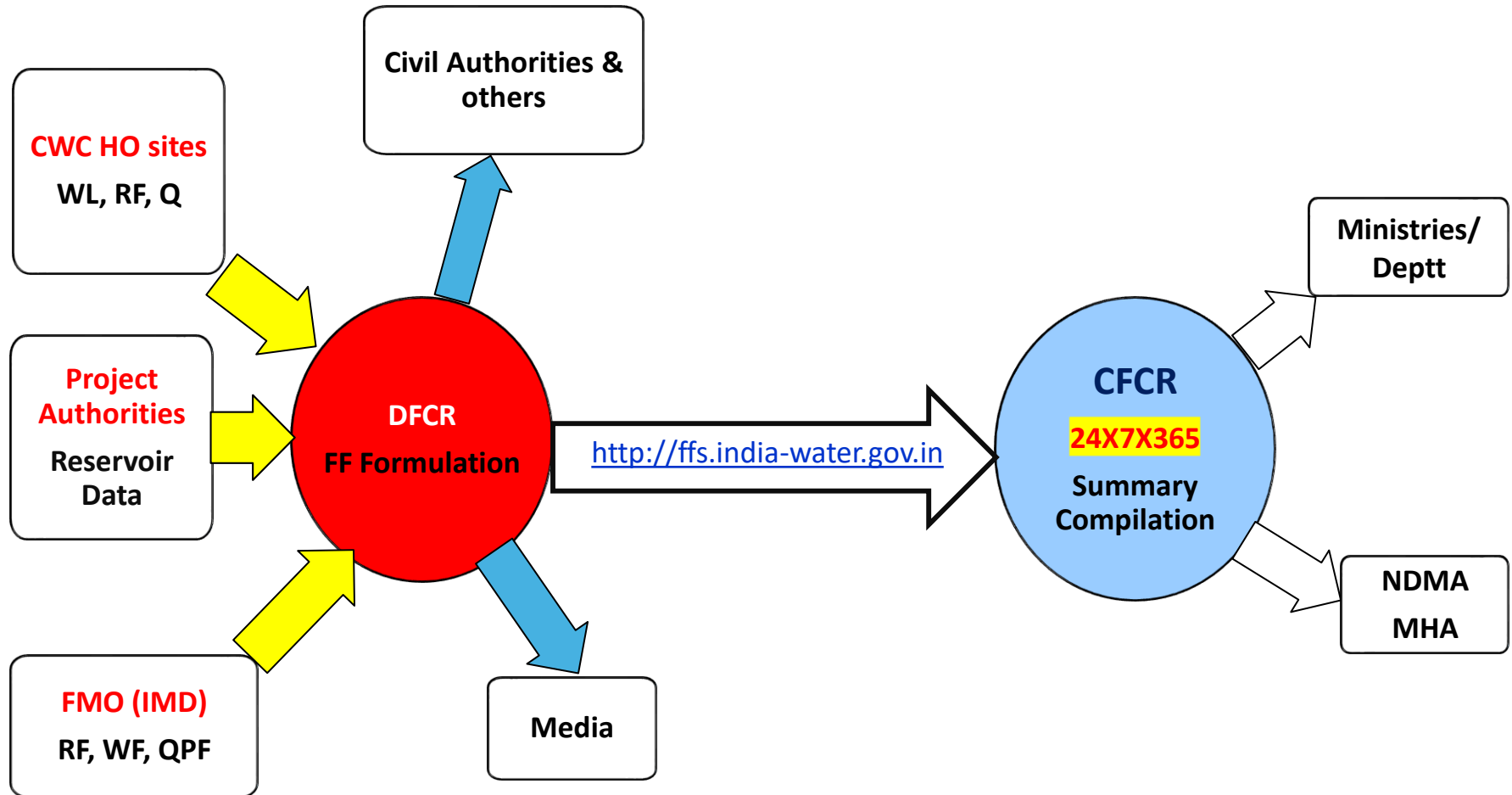
## Expansion of Flood Forecasting Network of CWC



## Flood Preparedness by CWC

# Flood Forecasting Setup

**1 Central and 36 Divisional Flood Control Rooms**



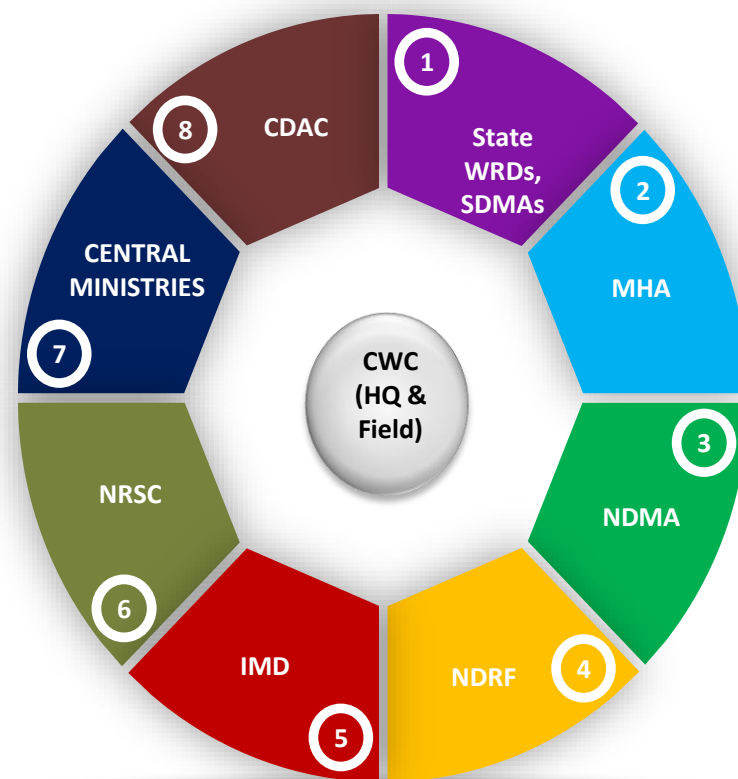
Coordination with other Agencies at Regional Level

24X7 Control Room of  
DoWR, RD & GR (MoJS)

# Flood Preparedness by CWC

## Institutional Framework

- Flood preparedness meeting with stakeholders including state agencies, SDMA and IMD officers before onset of monsoon
- One-Day Stakeholders' Workshop on CWC's Flood Forecasting Services on 09.02.2026 at CWC(HQ), New Delhi
- Conducting state-wise meeting to Strengthen Coordination and Sharing of Flood Forecasting Information between CWC and States agencies.
- Integration of states flood forecasting activities with C-Flood portal for 10 states.



# Flood Preparedness by CWC

## Flood Forecast Dissemination

- Email, Website, WhatsApp, Social media, Mobile App ,CAP Alert
- Flood Monitoring & Short Range Forecast: <https://ffs.india-water.gov.in/>
- 7-day Advisory Flood Forecast: <https://aff.india-water.gov.in/>
- Inundation Forecast: C-Floods <https://inf.cwc.gov.in/>
- CAP Alert  Sachet Integrated Alert System
- Flood Messages 
- CWCOfficial.FF 
- CWCOfficial\_FF 
- CWC Flood Updates 
- Mobile App FloodWatch India 





Download the "FloodWatch India" App!



केंद्रीय जल आयोग  
Central Water Commission  
जल शक्ति मंत्रालय  
Ministry of Jal Shakti  
भारत सरकार  
Government of India



## Flood Preparedness by CWC

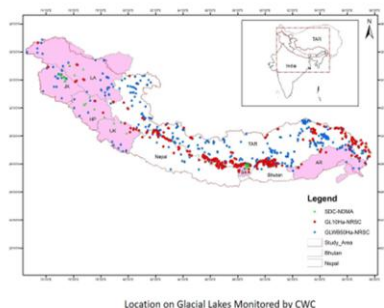
### Flood Forecast Dissemination

- ✓ Monitoring of Floods (<https://ffs.india-water.gov.in/>)
- ✓ Monitoring of Hydromet situation (IMD bulletins, various weather related websites, WRD websites)
- ✓ Issue of Daily, Red & Orange Flood Bulletins as per MHA SOP
- ✓ MHA Advisory (Daily)
- ✓ Daily Flood Situation report (Daily till Oct./Nov.) and Special Flood Advisory after SW Monsoon (As and when required)
- ✓ Issue of CAP Alerts to SDMA's through NDMA Sachet portal
- ✓ Monthly and Annual Flood Situation Report.
- ✓ Daily Flood updates through YouTube Channel (CWC Flood updates)
- ✓ Meetings with stakeholders on preparedness & updates



# Preparedness for GLOF

1. **Monitoring of 2843 Glacial Lakes/ Water Bodies** having size greater than 10 ha for period June-October every year using remote sensing technology.
2. **Monthly Monitoring Reports** are prepared and shared with disaster management authorities/State agencies/e-published in CWC website.
3. **Flow path mapping** of 100 Glacial Lakes located in India along with CWC HO stations and projects enroute, completed.
4. Finalization of Remote sensing-based **Criteria for Risk indexing of Glacial Lakes** in IHR. Risk assessment of 100 glacial lakes in India completed. 581 Glacial Lakes in progress.
5. **Guideline for Assessment of GLOF** specific to river valley projects in IHR and **Guideline for Structural measures** to mitigate adverse effect of GLOF on Dam/Barrages published in August 2025.
6. **46 HE Projects** in Himalayas identified vulnerable to GLOF. EWS installed in all projects.

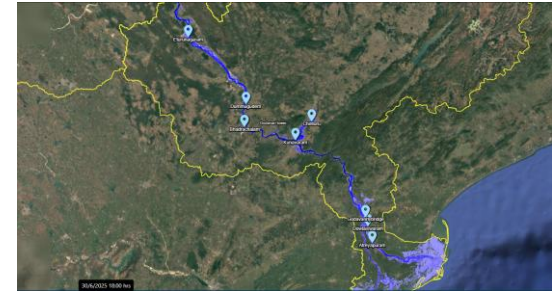
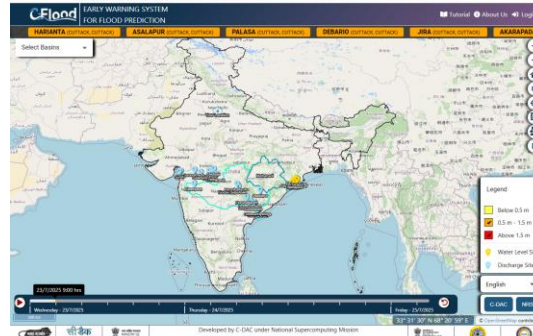


Covering 4 states ( Uttarakand ,HP ,  
Sikkim and Arunachal Pradesh) & 2 UTs  
(JK and Ladakh)

Himalayan Region  
+  
TAR region draining  
to India

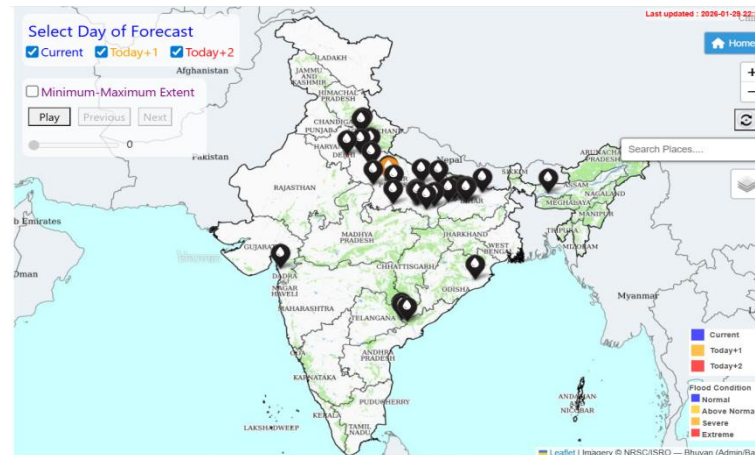
## Inundation Forecasting

- Godavari basin
- Tapi basin
- Mahanadi basin



- Unified Portal Launched by Hon'ble Union Minister, Jal Shakti on 2<sup>nd</sup> July 2025. Planned to **integrated station specific forecast in 2026**
- Libraries based inundation developed for 38 Stations as detailed below

- ✓ Andhra Pradesh: 1
- ✓ Assam: 1
- ✓ Bihar: 9
- ✓ Delhi: 1
- ✓ Gujarat: 1
- ✓ Odisha: 1
- ✓ Telangana: 2
- ✓ Uttar Pradesh: 19
- ✓ Uttarakhand: 2
- ✓ Haryana: 1



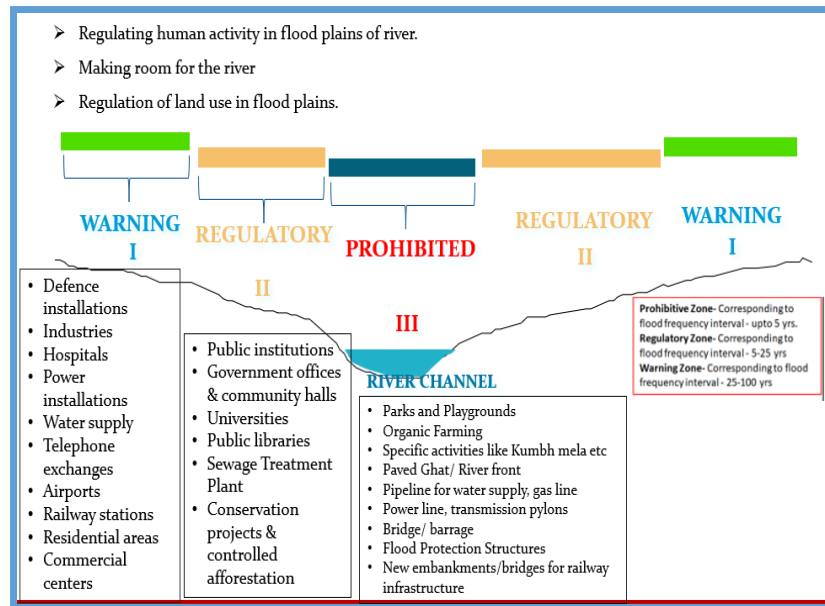
Planned to cover all the flood forecasting stations by 2027

# Flood Preparedness by CWC

## Flood Crisis Management Team (FCMT)

- Crisis Management Plan (CMP) 2025
  - 43(from 26) teams for 158 reservoirs (from 74) in 19 basins (from 8)
  - Involved Dam authorities, SDSOs & SDMAs, CWC, BBMB, NCA, DVC, GRMB, KRMB, GFCC
- State level Integrated reservoir operation agreed in Krishna Basin between Andhra Pradesh and Karnataka

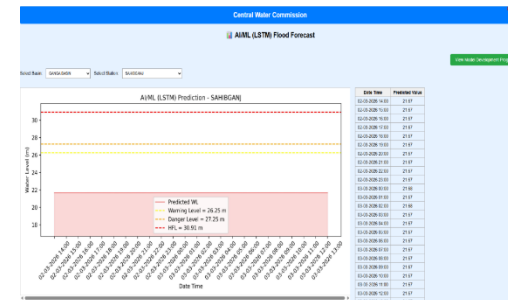
## Flood Plain Zoning



- Technical Guidelines on Model Flood plain Zoning circulated to States for implementation
- 8 states have adopted through Act/Policy

|                    |                                             |
|--------------------|---------------------------------------------|
| <u>Manipur</u>     | <u>Bihar</u>                                |
| <u>Rajasthan</u>   | <u>Arunachal Pradesh</u>                    |
| <u>J&amp;K</u>     | <u>Assam (Bodoland Territorial Council)</u> |
| <u>Uttarakhand</u> | <u>Tamil Nadu</u>                           |

## Implementation of AI/ML





## Co-operation requested from States

- ✓ Sharing of Near Real-time Hydrometeorological data of control structures with CWC through WIMS(NWIC/SWIC)
- ✓ Integrated Reservoir Operation
  - Co-operation through (FCMT ) - Flood Crisis Management Team
  - Utilization of DSS for Ganga Basin
- ✓ Realtime verification of Inundation forecast
- ✓ Implementation of Flood Plain Zoning
- ✓ Sharing of flood damage data for compilation
- ✓ Utilisation of flood Watch App amongst Govt officials & General population
- ✓ Fixation of Danger Level and Warning Level at H.O Sites in their administrative boundaries.
- ✓ URLs and API's of the flood forecasting portals/decision support systems being operated by the State Governments may kindly be shared with CWC. Further, for seamless exchange of real-time hydrological and meteorological data through an API-based mechanism, it is requested that a Nodal Officer from State organizations may kindly be nominated.



**THANK YOU**



Flood Forecasting Portal



Central Water Commission

Current Flood Forecast

Site

Hydrograph

Table View

7 Day Advisory Flood Forecast

DHARAMTUL (56.13 m) ↓

KARIMGANJ (15.25 m) ↓

DHUBRI (28.75 m) ↓

Search Stations

Base Stations

Level forecast station

Inflow forecast station

Flood Monitoring Station

Forecast available for Inflow forecast station

Above normal flood

Severe flood

Extreme flood

Afghanistan

LADAKH

YEMEN

SOMALIA

500 km

People's Republic of China

| District / Town | Basin       | River       | Division                                    | RL of Zero gauge |
|-----------------|-------------|-------------|---------------------------------------------|------------------|
| DHUBRI          | Brahmaputra | Brahmaputra | Middle Brahmaputra Division (MBD), Guwahati | 0                |

| Warning Level (WL) | Danger Level (DL) | Highest Flood Level (HFL) | HFL Attained Date |
|--------------------|-------------------|---------------------------|-------------------|
| 27.62              | 28.62             | 30.37                     | 2019-07-17        |

Present Water Level

(in meters)

28.75 ↓

Sat, 7, Jun 2025, 9:00 AM

Cumulative Daily Rainfall

(in millimeters)

0

Sat, 7, Jun 2025, 8:30 AM

Level Forecast (10)

(in meters)

28.69 ↓

Sat, 7, Jun 2025, 9:00 PM

Hydrograph view of - Dhubri

Forecasted Water Level

Measured Water Level

Water Level (m)

31

30

29

28

27

HFL

Danger

Warning

Jun 4, 2025, 12:30

Jun 4, 2025, 16:30

Jun 5, 2025, 00:30

Jun 5, 2025, 04:30

Jun 5, 2025, 08:30

Jun 6, 2025, 12:30

Jun 6, 2025, 16:30

Jun 6, 2025, 20:30

Jun 7, 2025, 00:30

Jun 7, 2025, 04:30

Jun 7, 2025, 08:30

Jun 7, 2025, 12:30

Jun 7, 2025, 16:30

Jun 7, 2025, 20:30

Date / Time

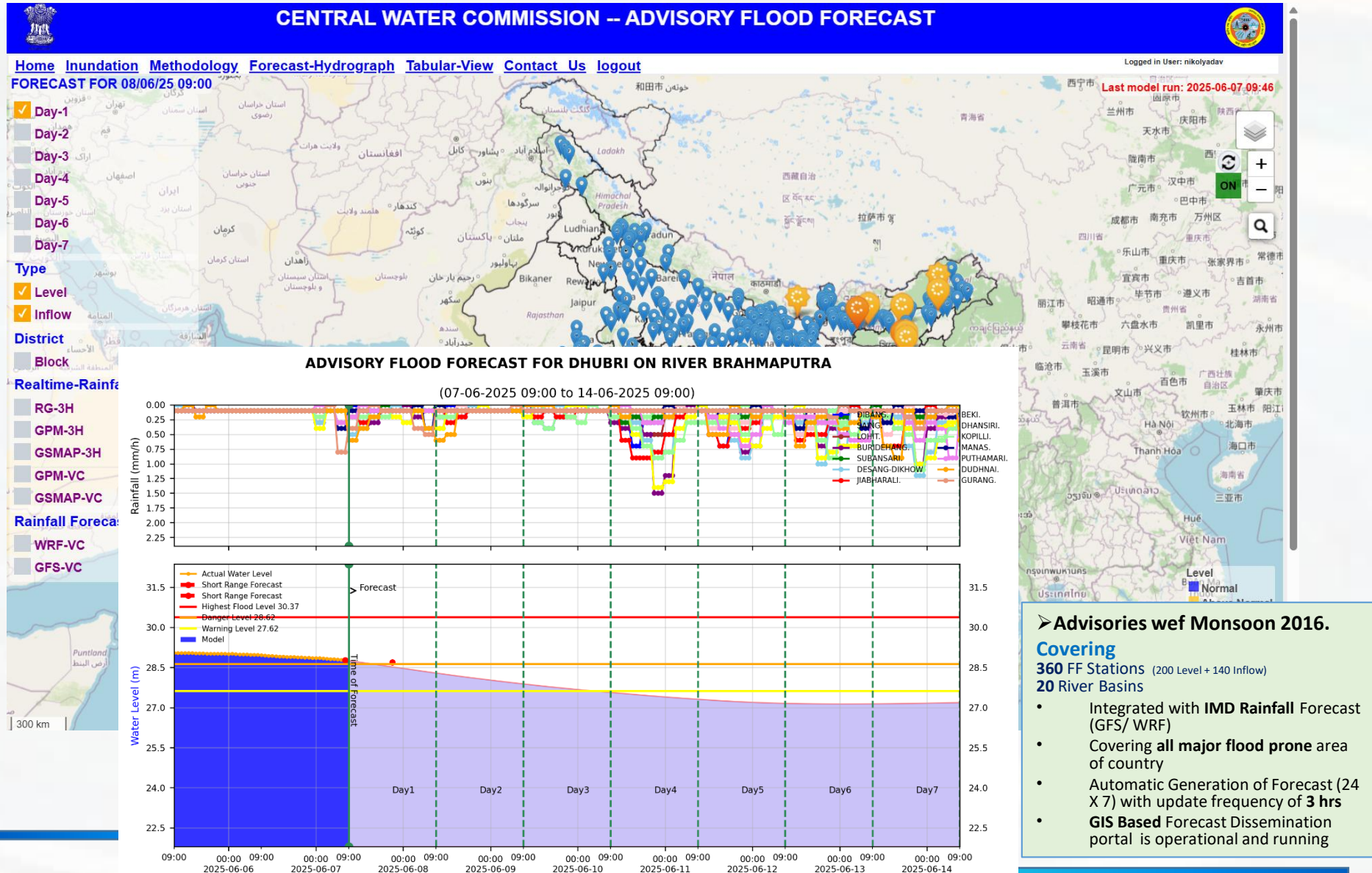
About Us

FloodWatch India

# Flood Preparedness by CWC

## 7-Day Advisory Forecast

<https://aff.india-water.gov.in/>



# Flood Preparedness by CWC

## Combined (CWC, IMD & NDRF) Reporting Format for MHA

| Ministry of Home Affairs<br>Flood Situation in sub-basins                                                                                                                              |                           |       |           |                    |       |       |       |       |                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------|-----------|--------------------|-------|-------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part- I                                                                                                                                                                                |                           |       |           |                    |       |       |       |       |                                                                                                                                                                                                                                                                                        |
| <div> <div>Highest Flood Level</div> <div>Danger Line</div> <div>Warning Line</div> </div> <div> <div>EXTREME</div> <div>SEVERE</div> <div>Above NORMAL</div> </div>                   |                           |       |           |                    |       |       |       |       |                                                                                                                                                                                                                                                                                        |
| <div> <div>OFF category (mm)</div> <div>1</div> <div>15.0</div> <div>15.5</div> <div>16.0</div> <div>16.5</div> <div>17.0</div> <div>17.5</div> <div>18.0</div> <div>18.5</div> </div> |                           |       |           |                    |       |       |       |       |                                                                                                                                                                                                                                                                                        |
| Sl. No.                                                                                                                                                                                | River/Sub-Basin/Basin     | State | District  | Rainfall Situation |       |       |       |       | Remarks/ Advisories                                                                                                                                                                                                                                                                    |
|                                                                                                                                                                                        |                           |       |           | Day 1              | Day 2 | Day 3 | Day 4 | Day 5 |                                                                                                                                                                                                                                                                                        |
| 1                                                                                                                                                                                      | Gandak/ Gandak/Ganga      | Bihar | Galaganj  |                    |       |       |       |       | I. First Meeting on the Monsoon Preparedness was conducted by Bihar State Disaster Management, Patna on 10.06.2021.<br>II. 11 teams available in Bihar i.e. Bettiah-2, Motihari-1, Gopalganj-1, Kishanganj-1, Darbhanga-1, Araria-1, Supaul-1 & Didarganj-3.                           |
| 2                                                                                                                                                                                      | Desang/Desang/Brahmaputra | Assam | Sivasagar |                    |       |       |       |       | I. First Meeting on the Monsoon Preparedness was held on 08.05.2021.<br>II. A review meeting on Monsoon Preparedness was held on 21.05.2021.<br>III. 10 teams available in Assam i.e. IWT Guwahati-1, Cachar-1, Barpeta-1, Bongaigaon-1, Jorhat-1, Sonitpur-1, Sivasagar-1 & Dhemaj-3. |

| Reservoirs Position                                                                                                                                                                                                                                                                                                                                                                    |                  |                                  |           |                                                             |                    |       |       |       |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------|-----------|-------------------------------------------------------------|--------------------|-------|-------|-------|-------|
| Part- II                                                                                                                                                                                                                                                                                                                                                                               |                  |                                  |           |                                                             |                    |       |       |       |       |
| <div> <div>Storage above 85%</div> <div>Storage above 60%</div> <div>Storage above 50%</div> </div> <div> <div>OFF category (mm)</div> <div>1</div> <div>15.0</div> <div>15.5</div> <div>16.0</div> <div>16.5</div> <div>17.0</div> <div>17.5</div> <div>18.0</div> <div>18.5</div> </div>                                                                                             |                  |                                  |           |                                                             |                    |       |       |       |       |
| #                                                                                                                                                                                                                                                                                                                                                                                      | Reservoir/ Dams  | River/ Sub-Basin / Basin         | State     | US/ DS District                                             | Rainfall situation |       |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                        |                  |                                  |           |                                                             | Day 1              | Day 2 | Day 3 | Day 4 | Day 5 |
| 1                                                                                                                                                                                                                                                                                                                                                                                      | Almatti Dam      | Krishna/ Upper Krishna/ Krishna  | Karnataka | Kolhapur (Maharashtra)/ Bagalkote (Karnataka)               |                    |       |       |       |       |
| 2                                                                                                                                                                                                                                                                                                                                                                                      | Panchet Dam      | Damodar East/ Ganga              | Jharkhand | Dhanbad                                                     |                    |       |       |       |       |
| 3                                                                                                                                                                                                                                                                                                                                                                                      | Bhavanisagar Dam | Bhavani/ Middle Cauvery/ Cauvery | Tamilnadu | Palakad (Kerala), Nilgiri (TN), Coimbatore (TN)/ Erode (TN) |                    |       |       |       |       |
| 4                                                                                                                                                                                                                                                                                                                                                                                      | Kabini           | Kabini/ Cauvery                  | Karnataka | Wayanad (Kerala)/ Mysuru (Karnataka)                        |                    |       |       |       |       |
| Remarks/ Advisories                                                                                                                                                                                                                                                                                                                                                                    |                  |                                  |           |                                                             |                    |       |       |       |       |
| • 5 teams available in Karnataka i.e. Bengaluru-1, Dakshin Kannada-1, Kodagu-1, Raichur-1 & Belagavi-1.                                                                                                                                                                                                                                                                                |                  |                                  |           |                                                             |                    |       |       |       |       |
| • 4 teams available in Jharkhand i.e. East Singhbhum (Jamshedpur)-1, Dogahtar-1 and Ranchi-2.                                                                                                                                                                                                                                                                                          |                  |                                  |           |                                                             |                    |       |       |       |       |
| • 1 team available at Chennai (Tamilnadu).                                                                                                                                                                                                                                                                                                                                             |                  |                                  |           |                                                             |                    |       |       |       |       |
| • Karnataka as mentioned above.                                                                                                                                                                                                                                                                                                                                                        |                  |                                  |           |                                                             |                    |       |       |       |       |
| • Kerala-<br>i. First Meeting on the Monsoon Preparedness was conducted by State Relief Commissioner, Kerala on 12.05.2021.<br>ii. A review meeting on Monsoon Preparedness was conducted by Chief Secretary, Kerala on 19.05.2021.<br>iii. 09 teams available i.e. Idukki-1, Kollam-1, Pattanamthitta-1, Alappuzha-1, Ernakulam-1, Thrissur-1, Malappuram-1, Kozhikode-1 & Wayanad-1. |                  |                                  |           |                                                             |                    |       |       |       |       |

Note: Based on above information, Project Authority may regulate the reservoirs as per standard operating manuals/ rule levels to avoid downstream flooding and upstream submergence.

## Finalized and Operational since July 2021

